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Project Title:	Simulated Microgravity Measurement Techniques for the Study of Dynamic Effects in Phospholipid Surfactants		
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Zip Code:	44106-1712	**Congressional District:**	11
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	We shall conduct a program of experiment and theory designed to provide rapid and reliable measurements of the dynamic properties of biologically important surfactants, such as pulmonary surfactants, under a wide variety of conditions. The essence of our technique rests on our ability to simulate a microgravity environment - indeed, we can simulate any environment between 0g and 1g, including those that vary with time - with a magnetic field so that we may perform resonance measurements on surfactant-coated liquid bridges. We will examine how the surface tension of an expanding or contracting air-water interface depends on the nature of the biological surfactants present and the rate at which the surface is expanding or contracting. Using our magnetic levitation technique, we shall undertake a second project that deals with fluid flow through randomly porous media. We shall examine how the rate of flow depends on time, particle size, and the total body force (corresponding to the difference between the magnetic force and gravity). This will allow us to pursue work in support of the lunar and Martian missions, for which we can simulate conditions of 0.17g and 0.39g, respectively.Connecting this		

Task Description:	study with our dynamic surface tension studies, we intend to examine the flow behavior in the presence of surfactant. Because existing methods of measurement of dynamic surface tension have significant limitations, we have developed a sensitive technique that depends on the measurement of the resonance frequency of an oscillating liquid bridge, which requires a microgravity environment. Using a strong magnetic field gradient to simulate this environment, we will oscillate the field around this simulated microgravity set-point to induce motion in the bridge. As the resonance behavior of the bridge is exquisitely sensitive to a number of physical parameters, we can accurately determine the surface tension of the bridge. Moreover, by controlling the amplitude of the magnetic field oscillations, we can extract dynamic surface tension information, viz., how the surface tension of the surfactant-coated liquid changes with the rate of change of the liquid's surface area. These measurements can be performed at frequencies ranging from d.c. to 50 Hz, providing a dynamic range far greater than is possible with alternative techniques, and facilitating a physical understanding of the underlying dynamic surfactant behavior. Overall, our results will provide an enhanced understanding of the physical mechanisms involved in biological control of surface tension. The objective of our new study of fluid flow through randomly-packed colloids in simulated reduced gravity is to support the lunar and Martian missions by modeling complex flow processes in appropriate gravitational environments. We will begin with monodisperse colloids, and successively add complexity by introducing polydispersity in both size and shape, ultimately simulating soils for agricultural purposes. Thus, with our technique, which facilitates very rapid changes in surface area, we wish to address several questions. What is the role of the minor lipids, such as PC, PG, and phosphatidylinositol in the process of DPPC adsorption in the absence of proteins? Specifically, how are the minor lipids "squeezed out" from the surface during the formation of the DPPC layer? On what time scales does this occur? [This question is particularly appropriate given the very large rate of change of area that we can generate in our system]. Other important questions include: How does overcompression affect the hysteresis behavior of the surface tension with area? It has been suggested that overcompression may result not only in single cycle hysteresis, but a general trend of increasing surface tension during subsequent cycling. We will therefore examine this issue not only in terms of the number of compression-expansion cycles, but also in terms of the frequency. Turning to fluid flow through randomly-packed colloids, our ultimate goal is to simulate fluid imbibition and flow through real soils in lunar and Martian gravitational conditions. Long space missions will require on-board life-support, including the ability to sustain agriculture in a less-than-1g environment. There are very few methods that allow us to simulate aqueous flow through soils in a reduced (but not zero) gravitational environment. Magnetic levitation is an ideal method, as it allows us to "dial in" a desired level of gravity and hold it for long periods of time. Additionally, we have the ability to vary the total body force on the fluid with time, facilitating investigation of methods to improve fluid transport. Along the way -- i.e., with less complex systems -- we expect to learn about effects of capillarity in reduced gravitational envirionments, including effects such as pinning and Haines' jumps.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our work on dynamic surface tension is related to medical issues such as respiritory distress syndrome in neonates, who often are unable to synthesize sufficient surfactant to facilitate pressure balance across the alveoli walls. We expect that our work will aid in understanding this problem, and provide information to create an appropriate cocktail of surfactants to alleviate this problem. Our work on fluid flow in reduced gravity is aimed primarily at extra-terrestrial agriculture, although it will help us understand phenomena such as fluid pinning, an issue of importance in fluid flow through colloids and solis on earth.
Task Progress:	Our recent work has turned to fluid flow through randomly-packed colloids in support of the new Lunar - Martian paradigm. We have performed a series of magnetic levitation measurements on a 3-dimensional porous medium consisting of a randomly-packed bed of monodisperse spheres, radius = 250 mm. The fluid interface height was measured as a function of time for levels of effective gravitational acceleration between 0g and 1g. We have obtained rise height vs. time results for the extreme cases, viz., 0g and 1g, as well as for 0.5g. The data reveal several interesting features. First, the rate of rise initially is largest for the 0g environment, followed by the 0.5g and 1g environments. In the short time limit the data were found to obey the Washburn equation, which involves the average capillary pressure, relatively well. The characteristic time tw associated with the initial Washburn-like rise is of order 30 s for the 0g case, about 20 s for the 1g case, and only a few seconds for the 0.5g case; beyond these times the data deviate from the Washburn form. We do not yet understand why tw, which is related to the equilibrium rise height as t goes to infinity, appears to be non-monotonic with g. For the 1g case, at longer times the rise does not flatten out, and in fact the rate surpasses the zero gravity case. Even more peculiar is the case of 0.5g, which shows a significant slowing down after the very early (and short-lived) Washburn-like rise. Again, there is an apparent non-monotonicity for the longer time behavior. Additionally, in all cases the fluid front becomes pinned at several different heights, each time remaining fixed for tens of seconds before spontaneously depinning. These effects were examined experimentally in 1g by Delker, et al, who analyzed their data in terms of models for interface pinning in random media. They found that beyond tw the interface advances by small jumps, fully wetting one or more beads at a time before moving on. Their results, however, offer no clues as to our observed behavior wherein the pinning appears to occur more regularly for the case of 0g than for either 0.5g or 1g. Preliminary work also has been performed using smaller, randomly-packed spherical beads, indicating qualitatively similar results, but with quantitative differences. We also have created a regular simple-cubic lattice structure of spherical glass beads by means of sintering. Work is currently underway to understand the effects of packing, and whether a regular periodic colloidal lattice influences the transport rate and pinning.
Bibliography Type:	Description: (Last Updated: 12/04/2024)
Dissertations and Theses	Patel NM. "Electrooptic Studies of Liquid Crystalline Phases and Magnetically Levitated Aqueous Bridges." Ph.D. thesis, Physics Dept, Case Western Reserve U, March 2004. , Mar-2004

Presentation	C. Rosenblatt "Dancing Fluids in Simulated Microgravity" Colloquium, Dept. of Physics, University of Manchester, UK Sep-2003
Presentation	C. Rosenblatt "Dancing Fluids in Simulated Microgravity" Colloquium, Dept. of Physics, University of Pennsylvania Sep-2003
Presentation	C. Rosenblatt "Dancing Fluids in Simulated Microgravity" Colloquium, Dept. of Physics, Brown University Feb-2004
Presentation	C. Rosenblatt "Dancing Fluids in Simulated Microgravity" Colloquium, Dept. of Physics, Pennsylvania State University May-2004
Presentation	C. Rosenblatt "Measurements of Surfactant Squeeze-out Using Magnetically-Levitated Liquid Bridges " NASA/JPL Fundamental Physics Workshop, Solvang, CA Apr-2004